

Solution to Ramanujan Challenge Problem 2.2: Euler's Constant γ as an Apéry Limit

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Abstract

We prove unconditionally that the quotient defined by the four-term recurrence in Problem 2.2 converges to Euler's constant. The challenge recurrence is an exact first-order Ore transform of a recurrence with positive hypergeometric solutions. For the latter, the quotient is a weighted average of harmonic numbers. A finite birth-death (Stein) identity gives a quantitative second moment bound at the cubic saddle $k^3 = (n - k)^2$; elementary harmonic-number bounds then imply convergence to γ . Thus no asymptotic theorem for a resonant recurrence and no cited analytic limit is required.

1 The challenge recurrence

After shifting the three negative initial indices, write the challenge solutions as $P_n = p_{n-3}$ and $Q_n = q_{n-3}$. Thus

$$(P_0, P_1, P_2) = (0, 7, 179), \quad (Q_0, Q_1, Q_2) = (1, 12, 306).$$

They satisfy

$$c_0(n)U_{n+3} + c_1(n)U_{n+2} + c_2(n)U_{n+1} + c_3(n)U_n = 0, \tag{1}$$

where

$$\begin{aligned} c_0(n) &= -8n^3 - 51n^2 - 105n - 68, \\ c_1(n) &= 24n^5 + 337n^4 + 1833n^3 + 4818n^2 + 6092n + 2928, \\ c_2(n) &= -(n+2)(n+3)(24n^5 + 273n^4 + 1150n^3 + 2154n^2 + 1635n + 268), \\ c_3(n) &= (n+1)(n+2)^4(n+3)(8n^3 + 75n^2 + 231n + 232). \end{aligned}$$

We will prove that $P_n/Q_n \rightarrow \gamma$. Shifting the index back then gives the limit in the statement of the problem.

2 A positive hypergeometric model

Let A_n, B_n be the two solutions of

$$\begin{aligned} &(n+3)^2(8n+11)(8n+19)R_{n+3} \\ &\quad - (n+3)(8n+11)(24n^2 + 145n + 215)R_{n+2} \\ &\quad + (8n+27)(24n^3 + 105n^2 + 124n + 25)R_{n+1} \\ &\quad - (n+2)^2(8n+19)(8n+27)R_n = 0, \end{aligned} \tag{2}$$

with

$$(A_0, A_1, A_2) = (-1, 4, 77/4), \quad (B_0, B_1, B_2) = (1, 7, 65/2).$$

Put $H_m = \sum_{j=1}^m 1/j$, with $H_0 = 0$, and

$$w_{n,k} = \frac{\binom{n}{k}^2}{k!} \quad (0 \leq k \leq n).$$

Proposition 1 (finite-sum formulas). *For every $n \geq 0$,*

$$B_n = \sum_{k=0}^n (2n + k + 1) w_{n,k}, \quad (3)$$

$$A_n = \sum_{k=0}^n w_{n,k} ((2n + k + 1)(3H_k - 2H_{n-k}) - 1). \quad (4)$$

In particular, $B_n > 0$.

Proof. The adjacent quotient

$$\frac{w_{n,k+1}}{w_{n,k}} = \frac{(n-k)^2}{(k+1)^3}$$

makes both summands hypergeometric (with the harmonic part handled by $H_{k+1} - H_k = 1/(k+1)$). Substitution in (2), followed by clearing denominators, is a telescoping identity in k ; the boundary terms vanish at $k = 0$ and $k = n+1$. The resulting sums have the displayed initial triples, so uniqueness for (2) proves the formulas. This finite calculation, including the two telescoping certificates and their boundary terms, is checked symbolically and in Lean in `RamanujanChallenge/Problem22.lean`. $\square$

Let $\tilde{A}_n = (n!)^2 A_n$ and $\tilde{B}_n = (n!)^2 B_n$. Define the first-order operator

$$\mathcal{O}(T)_n = \frac{T_{n+1} + (n+1)(3n+4)T_n}{8n+11}. \quad (5)$$

Proposition 2 (exact Ore transform). *The operator (5) sends every factorial-scaled solution of (2) to a solution of (1), and*

$$P_n = \mathcal{O}(\tilde{A})_n, \text{quad} Q_n = \mathcal{O}(\tilde{B})_n.$$

Moreover $Q_n > 0$ for all n .

Proof. Insert (5) into the left side of (1). After clearing denominators, it is a polynomial linear combination of (2) at n and at $n+1$. The transformed initial triples are respectively $(0, 7, 179)$ and $(1, 12, 306)$, proving the identities by uniqueness. Positivity follows from (3): both terms in the numerator of (5) are positive. The polynomial identity is also machine-checked in the Lean source cited above. $\square$

It remains to prove $A_n/B_n \rightarrow \gamma$. Indeed, if $a_n = (n+1)(3n+4)$, then

$$\frac{\mathcal{O}(\tilde{A})_n}{\mathcal{O}(\tilde{B})_n} = \frac{\tilde{A}_{n+1} + a_n \tilde{A}_n}{\tilde{B}_{n+1} + a_n \tilde{B}_n}.$$

This is a positive weighted average of $\tilde{A}_{n+1}/\tilde{B}_{n+1}$ and $\tilde{A}_n/\tilde{B}_n$, so it has the same limit.

3 A finite Stein identity

Use the positive weights

$$W_{n,k} = (2n + k + 1)w_{n,k}, \quad Z_n = \sum_{k=0}^n W_{n,k} = B_n,$$

and let K be the random variable on $\{0, \dots, n\}$ with $\Pr(K = k) = W_{n,k}/Z_n$. Define

$$\begin{aligned} b_n(k) &= (2n + k + 2)(n - k)^2, \\ d_n(k) &= (2n + k)k^3, \\ G_n(k) &= k^3 - (n - k)^2. \end{aligned}$$

The adjacent-weight identity is exactly

$$W_{n,k}b_n(k) = W_{n,k+1}d_n(k+1). \quad (6)$$

Since $b_n(n) = d_n(0) = 0$, reindexing the finite sum gives, for every function f ,

$$\mathbb{E}[b_n(K)(f(K+1) - f(K)) + (b_n(K) - d_n(K))f(K)] = 0. \quad (7)$$

Two elementary polynomial identities drive the estimate:

$$G_n(k+1) - G_n(k) = 3k^2 + k + 2n, \quad (8)$$

$$b_n(k) - d_n(k) = 2(n - k)^2 - (2n + k)G_n(k). \quad (9)$$

Lemma 3 (moment bounds). *For $n \geq 1$,*

$$\mathbb{E}[K^3] \leq \frac{5}{2}n^2, \quad (10)$$

$$\mathbb{E}[K^2] \leq \frac{7}{2}n\sqrt{n}, \quad (11)$$

$$\mathbb{E}[G_n(K)^2] \leq 81n^3\sqrt{n}. \quad (12)$$

Proof. Taking $f = 1$ in (7) gives $\mathbb{E}[b_n(K)] = \mathbb{E}[d_n(K)]$. On the support,

$$d_n(k) \geq 2nk^3, \quad b_n(k) \leq 5n^3,$$

which proves (10). The interpolation inequality

$$x^2 \leq n + \frac{x^3}{\sqrt{n}} \quad (x \geq 0)$$

then gives (11).

Next take $f = G_n$ in (7) and use (8)–(9). We obtain the exact identity

$$\mathbb{E}[(2n + K)G_n(K)^2] = \mathbb{E}[2(n - K)^2G_n(K) + b_n(K)(3K^2 + K + 2n)].$$

Young's inequality and the support bounds give

$$\begin{aligned} 2(n - k)^2G_n(k) &\leq nG_n(k)^2 + n^3, \\ b_n(k)(3k^2 + k + 2n) &\leq 20n^3k^2 + 10n^4. \end{aligned}$$

The left side is at least $2n\mathbb{E}[G_n(K)^2]$. Hence

$$n\mathbb{E}[G_n(K)^2] \leq 20n^3\mathbb{E}[K^2] + 11n^4 \leq 81n^4\sqrt{n},$$

which proves (12). □

4 From the cubic saddle to harmonic numbers

The quotient in Proposition 1 is

$$\frac{A_n}{B_n} = \mathbb{E} \left[3H_K - 2H_{n-K} - \frac{1}{2n+K+1} \right]. \quad (13)$$

We first omit the last, uniformly negligible, term.

For $m \geq 1$, the standard elementary bounds for harmonic numbers imply

$$0 \leq H_m - \log m - \gamma \leq \frac{1}{m}. \quad (14)$$

Fix $0 < \delta \leq 1/16$ and call k good when

$$|G_n(k)| \leq \delta n^2.$$

If $n \geq 16$ and k is good, direct comparison in $k^3 = (n-k)^2 + G_n(k)$ gives

$$1 \leq k < n, \text{ good } k^3 \geq \frac{3}{16}n^2, \text{ good } n-k \geq \frac{n}{2}. \quad (15)$$

Suppose also that $M \geq 1$ and $16M^3 \leq 3n^2$. Then (15) implies $k \geq M$ and $n-k \geq M$. Put

$$x = \frac{k^3}{(n-k)^2}.$$

The last inequality in (15) gives

$$|x - 1| = \frac{|G_n(k)|}{(n-k)^2} \leq 4\delta \leq \frac{1}{4}.$$

Since $|\log x| \leq 2|x - 1|$ in this range, (14) shows that every good k satisfies

$$|3H_k - 2H_{n-k} - \gamma| \leq \frac{5}{M} + 8\delta. \quad (16)$$

On the bad set we use only the coarse estimate

$$|3H_k - 2H_{n-k} - \gamma| \leq 6 + 5 \log n \quad (0 \leq k \leq n), \quad (17)$$

which follows from $0 \leq H_m \leq 1 + \log n$ and $0 < \gamma < 1$. Badness also implies

$$1 \leq \frac{G_n(k)^2}{\delta^2 n^4}.$$

Combining (12), (16), and (17) yields the quantitative estimate

$$\mathbb{E}[|3H_K - 2H_{n-K} - \gamma|] \leq \frac{5}{M} + 8\delta + \frac{81(6 + 5 \log n)}{\delta^2 \sqrt{n}}. \quad (18)$$

Given $\varepsilon > 0$, first choose $0 < \delta \leq \min(1/16, \varepsilon/32)$, then choose $M > 20/\varepsilon$. For all sufficiently large n the condition $16M^3 \leq 3n^2$ holds and the last term of (18) is less than $\varepsilon/2$. Therefore the left side tends to zero. Finally,

$$0 < \mathbb{E} \left[\frac{1}{2n+K+1} \right] \leq \frac{1}{2n+1} \longrightarrow 0.$$

Equation (13) now proves

$$\frac{A_n}{B_n} \longrightarrow \gamma. \quad (19)$$

5 Conclusion

By Proposition 2, the challenge quotient is a positive weighted average of two adjacent quotients in (19). Consequently

$$\lim_{n \rightarrow \infty} \frac{p_n}{q_n} = \gamma.$$

The complete argument has been formalized in Lean 4, in namespace `RamanujanChallenge.P22`. The public theorem is `problem22_solved`. Its axiom audit contains only the three standard Lean foundations `propext`, `Classical.choice`, and `Quot.sound`. The proof contains no `sorry` declarations or problem-specific axioms.